UNIT-11

p-BLOCK ELEMENTS

Group-13 Elements

• **Inert pair effect:**It is the reluctance of the *s*-electrons of the valence shell to take part in bonding. It arises due to poor or ineffective shielding of the ns^2 Electrons of the valence shell by the intervening *d*- and/or *f*-electrons. Inert pair effect increases down a group and thus the elements present in the lower part of the group show lower oxidation states which is two units less than the highest group oxidation state. In group 13 elements, inert pair effect is more pronounced in Tl.

• **Oxidation states:**B and Al show an oxidation state of +3 only while Indium and thallium show oxidation states of both +1 and +3. Further due to inert pair effect, as we move down the group, the stability of +3 oxidation state decreases while that of +1 oxidation state increases. For Tl, it is the +1 Oxidation state which is more stable than +3 as shown by the redox potential data :

$Tl^{3+}(aq) + 2e^- \rightarrow Tl^+(aq)$; $E° = +1.25$ V

The decrease in stability of +3 oxidation state with increasing atomic number is due to decrease in bond energy with increase in size from Al to Tl. As a result, the energy required to unpair the ns^2electrons is not compensated by the energy released in forming two additional bonds.Stability of +1 oxidation state follows the trend : Ga< In <Tl

• **Halides :** All the elements of group 13 (except thallium which preferably Forms Thallous monohalides) form trihalides of the general formula MX Where X = F, Cl, Br and I.

All the boron trihalides, *i.e.*, BF_3, BCl_3, $BBr_3$and BIand aluminumtrihalides *i.e.*, $AlCl_3$, $AlBr_3$and AlI_3(AlF_3being ionic) are covalentcompounds. Whereas boron trihalides exist as only monomers, aluminumtrihalides exist as dimers.

• **Nature of Trihalides:**The trihalides of group 13 elements have only six Electrons in the valence shell and hence have a strong tendency to accept a Pair of electrons to acquire the nearest inert gas configuration. Consequently Trihalides of group 13 elements behave as Lewis acids.

Group-14 Elements

• **Oxidation states:**All the elements show an oxidation state of +4. However, as we move down the group from C to Pb, the stability of +4 oxidation state Decrease while that of +2 oxidation state increases due to inert pair effect. Thus, +4 oxidation state is important for C and Si while +2 state becomes Increasingly important for Ge, Sn and Pb. The stability of the divalent state increases in the sequence Ge<Sn<Pb.

• **Catenation:**The property of self-linking ofan atoms of an element through Covalent bonds to form straight or branched chains and rings of different sizes is called **catenation**. The tendency of an element to form chains depends upon the strength of the element-element bond. Amongst the group 14

Elements, C-C bond strength (348 KJ mo1^{-1}) is maximum, therefore, carbon Shows maximum tendency for catenation. As we move down the group, the element-element bond strength decreases very rapidly,

i.e., Element-Element	C -C	Si-Si	Ge-Ge	Sn-Sn
Bond strength (kJ mol^{-1})	348	297	260	240

Therefore, the tendency for catenation decreases in the order C >>> Si > Ge~Sn>>Pb. Lead however, do not show any tendency for catenation.

***p*π-*p*π and *p*π-*d*π multiple bonding :** Amongst group 14 elements, carbon shows a pronounced ability to form *p*π-*p*π multiple bonds with itself (*e.g.*, in graphite) and with other elements especially nitrogen and oxygen. Silicon and other heavier elements of this group show negligible tendency of this type. This is the reason that silicon exists only in the diamond structure.

*p*π-*d*π multiple bonding has been observed particularly in case of silicon linked to oxygen and nitrogen. Thus trimethylamine, $N(CH_3)_3$is pyramidal (N is sp^3-hybridized) and is more basic whereas similar silicon compound trisilylamine,$N(SiH_3)_3$ is planar (N is sp^2-hybridized) and is less basic. In this case, the lonepair of nitrogen atom is transferred to the empty *d*-orbital of silicon (*p*π-*d*πoverlapping) leading to the planar structure.

• Chemical Properties

Nature of bonding: The compounds of group 14 elements which show an Oxidationstate of +4 is covalent while those which show an oxidation of +2 are Ionic in nature. Further, as we move down the group, the tendency of the elements to form covalent compounds decreases but the tendency to form ionic compounds increases.

Formation of halides:

(i) Tetra halides: All the elements of group 14 form tetra halides of the general formula, MXwhere X = F, Cl, Br or I. All these tetrahalides are covalent Compounds and have tetrahedral structures. The stability of these tetrahalides decreases as we move from C to Pb, *i.e.* CCl_4>$Sicl_4$>$Gecl_4$> $SnCl_4$> $PbCl_4$

CX_4>SiX_4>GeX_4>SnX_4> PbX_4

(ii) Dihalides : All the elements of group 14 form dihalides of the formula, MX_2 where X = F, Cl, Br or I. The stability of these dihalides increases as we move down the group from C to Pb. Thus, the dihalides of tin and Pb, *i.e.*, $SnCl_2$ and $PbCl_2$are quite stable. Dihalides are generally ionic in nature and behave as reducing agents, the reducing character (of M^{2+}species) decreases in the order

e.g., $GeCl_2$> $SnCl_2$> $PbCl_2$.

• **Formation of oxides :** All the elements of this group form two types of oxides : (i) monoxides and (ii) dioxide.

(i) Monoxides : All the elements of group 14 form monoxides of the general formula MO, *i.e.*, CO, SiO, GeO, SnO and PbO. These oxides except SiO and GeO are quite stable. CO is neutral while SnO and GeO are amphoteric.

(ii) Dioxides : All these elements form dioxides of the general formula, MO_2 *i.e.*, CO_2, SiO_2, GeO_2, SnO_2 and PbO. All these dioxides can be prepared by heating the elements in oxygen except lead which forms lead monoxide. CO_2 is a monomeric, linear molecule and hence exists as a gas while the dioxides of all other elements are crystalline solids with high melting points due to the formation of three dimentional network of bond.

• **Silicones :** Silicones are synthetic organosilicon compounds containing repeated R_2SiO units held by Si-O- Si linkages. Since the empirical formula R_2SiO (where R = CH_3 or C_6H_5 group) is similar to that of a ketone (R_2CO), the name silicones have been given to these materials.Short chains silicones are oily liquids, medium chains silicones behave asviscous oils, jellies and greases and those with very long chains behave as rubberyelastomers and resins.

Silicones are water repellent, heat resistant, chemically inert, resistant to oxidation and attack by organic acids and good electrical insulators. Silicones are used for making water proof papers, wool, textiles, wood, etc., as lubricants at high and low temperatures, as antifoaming agents, as sealants.Being biocompatible they are also used in surgical and cosmetic plants.

SOME IMPORTANT QUESTIONS :

1.What is general electronic configuration of the p – block elements?

Ans.ns^2np^{1-6}

2.Why is Boron metalloid?

Ans. Boron resembles with both metals and non- metals, therefore, it is metalloid.

3. Name two important ores of Boron.

Ans. (i) Borax,$Na_2B_4O_7.10H_2O$ (ii)Kernite ,$Na_2B_4O_7.2H_2O$

4. Name two isotopes of Boron.

Ans. ^{10}B and ^{11}B are isotopes of Boron.

5. Name two allotropes of Boron.

Ans. Amorphousand Crystalline.

6.Why does Boron not form B^{3+}ion?

Ans.Due to smallest atomic size and highest ionization enthalpy, Boron doesnot form B^{3+} ion.

7. Why is Boron used in nuclear reactor?

Ans.Boron can absorb neutrons.

8.Why does Boron resemble Si?

Ans.Both have similar charge over radius ratio.i.e. Similar polarizing power.

9. Why does Boron form stable electron deficient compound?

Ans. Boron has three valenceelectron, it will share three electrons other elements to form electron deficient compounds which are stable.

10. Why is boric acid (H_3BO_3) monobasic acid?

Ans.It accept a pair of electrons from OH^- ion of H_2O therefore, it is monobasic acid.

11. What typeof glass is obtained when borax is added?

Ans.Pyrex glass which is heat resistant is obtained when borax is added.

12. Why BH_3 exists in form of diborane?

Ans. BH_3 exists in form of diborane (B_2H_6) because it is electron deficient.

13. Why does BF_6^{3-}not exist?

Ans.It is because Boron does not have vacant d – orbitals.

14. Why do Boron halides form addition compounds with ammonia and amines?

Ans.It is because they are electron deficient.

15. In which pure form does carbon exist in nature?

Ans.Fullerene.

16. What is the hybridization of carbon indiamond?

Ans. sp^3

17. Give structure of CO_2?

Ans. O=C=O,It has linear structure.

18. What is producer gas?

Ans. Producer gas is mixture of CO and N_2 in the ratio of 2:1.

19. What is state of hybridization of C in CO_3^{2-} ?

Ans. sp^2 hybridization.

20. How does BF_3 act as catalyst in industrial process?

Ans. BF_3 is electron deficient; therefore act as catalyst in industrial process.

21. Why is BF_3 weaker Lewis acid than BCl_3 ?

Ans. BF_3 weaker Lewis acid than BCl_3 because of more effective back bonding in case of F due to smaller size than Cl .

22. Write the state of hybridization in BF_3?

Ans.B in BF_3 is sp^2 hybridization.

23. Why does boric acid act as Lewis acid?

Ans. It is because in boric acid, boron does not have its octet complete. It accepts OH^- from water in aqueous solution

$B(OH)_3 + H_2O \rightarrow [B(OH)_4]^- + H^+$

24. Why does boron form electron deficient compounds?

Ans.It is because boron has three valence electrons .It shares three electrons and gets six electrons i.e. its octet is not complete. Therefore, it forms electron deficient compounds.

25. Why does boron shows anomalous behavior in group 13 of the periodic table?

Ans.It is because boron has small size, high ionization enthalpy, absence of d- orbitals and high charge / size ratio.

26. Why does carbon exhibit catenation much more than any other element ?

Ans. Carbon exhibit catenation to maximum extent because of its small size, it can form stable covalent bonds with other carbon atoms.

27. CO2 is gas while SiO_2 is solid at room temperature. State one reason.

Ans. CO_2 exists as discrete molecules, therefore, has weak Vander Waals forces of attraction whereas SiO_2 is three dimensional network solid.

28. What is general formula of silicones?

Ans.R_2SiO is general formula of silicones.

29. What is inert pair effect?

Ans.The pair of electrons in valence s- orbital is reluctant to take part in bond formation due to poor shielding effect of d and f- orbitals in heavier elements is called inert pair effect.

30.B to Tl ,+1 oxidation state becomes more stable ,why ?

Ans.It is due to inert pair effect .

31. What are silicates ?

Ans. Silicates are minerals which consist of SiO_4^{4-} units arranged in different ways.

32. BCl_3 exists but BH_3 does not. Explain.

Ans.In BCl_3, Cl donates lone pair of electronsto vacant p- orbital of boron (back bonding) making it more stable whereas in BH_3 back bonding is not possible, therefore, it exists as dimer. Secondly, in BCl_3, Chlorine being bigger in size cannot form bridged bonds.

33. Silane gets hydrolyzed by water whereas alkane doesnot. Why?

Ans. Silane gets hydrolyzed because Silicon has vacant d- orbital whereas alkane cannot due to absence of d-orbital in Carbon.

34. SiCl4 gets hydrolyzed, CCl_4doesnot,why?

Ans.Si has vacant d- orbital whereas carbon does not have vacant d- orbital in Carbon.

35.$PbCl_4$ exists but $PbBr_4$, PbI_4 do not .Explain.

Ans. Chlorine is strong oxidizing agent than Bromine and iodine. ItoxidizesPb to higher oxidation state. Therefore, $PbCl_4$ existsbut PbBr4, PbI_4 does not. Secondly, bromine and iodine are larger in size.

36. Why is $PbCl_4$ good oxidizing agent?

Ans. $PbCl_4$ good oxidizing agent because Pb^{2+} is more stable than Pb^{4+} due to inert pair effect , Therefore, Pb^{4+} readily changes into Pb^{2+} by gaining two electrons.

37. Write the formula of inorganic benzene?

Ans. $B_3N_3H_6$

NCERT TEXT BOOK QUESTIONS :

38. Discuss the pattern of variation in the oxidation states of

(i) B to Tl and (ii) C to Pb

Ans. (i) Tendency to show +1 oxidation state increases down the group due to inert pair effect.

(ii) Tendency to show +2 oxidation state increases down the group due to inert pair effect.

39.How can you explain higher stability of BCl_3 as compared to $TlCl_3$?

Ans.It is because Tl^+ is more stable than Tl^{3+} due to inert pair effect hence $TlCl_3$ is less stable than BCl_3.

40. Why does boron triflouride behave as a Lewis acid?

Ans. It is because octet of boron is not complete, it is electron deficient Therefore, act as a Lewis acid.

41. Consider thecompound, BCl_3 and CCl_4.How will they behave with water? Justify.

Ans. BCl_3 is covalent and electron deficient hence it react with water.

$BCl_3 + 3H_2O \longrightarrow H_3BO_3 + 3HCl$

CCl_4 does not react with water because carbon does not have vacant d- orbital and octet of carbon is complete.

42.Is boric acid a protic acid ? Explain .

Ans. No boric acid aprotic acid .It is Lewis acid accepts OH^-.

$B(OH)_3 + H_2O \longrightarrow [B(OH)_4]^- + H^+$

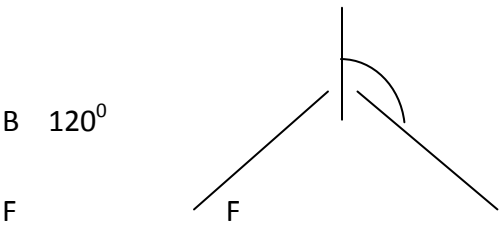
43. Explain what happens when boric acid is heated.

Ans. $H_3BO_3 \longrightarrow HBO_2 + H_2O$

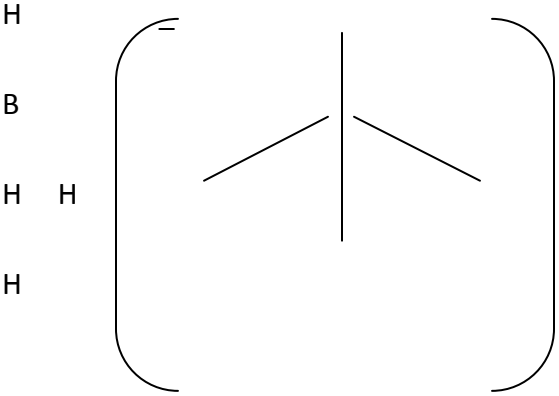
$HBO_2 \longrightarrow B_2O_3 + H_2O$

44.Describe the shapes of BF_3 and BH_4^- .Assign the hybridization of boron in these species.

Ans. BF_3 is trigonal planer due to sp^2 hybridization.



BH_4^- is tetrahedral due to sp^3 hybridization.



45.Write reactions to justify amphoteric nature of Aluminium.

Ans. $2Al + 6HCl \longrightarrow 2AlCl_3 + 3H_2$

$2Al + NaOH + 2H_2O \longrightarrow 2NaAlO_2 + 3H_2$

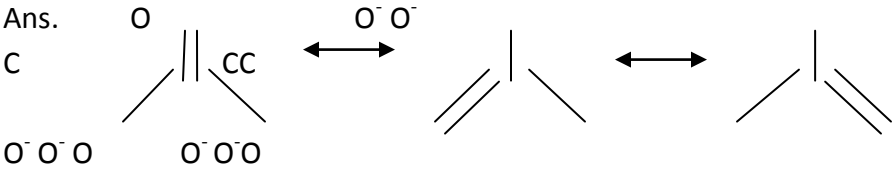
46.What are electron deficient compounds ? Are BCl_3 and $SiCl_4$ electron deficient species ? Explain .

Ans. Electron deficient compounds are those compounds in which octet of central atom is not complete.

BCl_3 is electron deficient compounds because its octet is not complete. $SiCl_4$ is not electron deficient compounds because its octet is complete.

47.Write the resonance structure of CO_3^{2-}.

Ans.



48.What is the state of hybridization of carbon in (a) CO_3^{2-} (b) diamond (c) graphite ?

Ans. (a) CO_3^{2-} - sp^2

(b) diamond - sp^3

(c) graphite - sp^2

49. Explain the difference in properties of graphite and diamond on the basis of their structures.

Ans.(i)Diamond is hard but graphite is soft.

(ii)Diamond is insulator but graphite is good conductor of heat and electricity.

50.Rationalise the given statement and give chemical reactions :

-Lead (II)Chloride reacts with Cl_2 to give $PbCl_4$.

- Lead (IV)Chloride is highly unstable towards heat.

-Lead is known not to form an iodide, PbI_4.

Ans.$PbCl_2 + Cl_2$ no reaction.

- It is because Pb can show +2 oxidation state more easily than +4 due to inert pair effect.

- $PbCl_4 \xrightarrow{heat} PbCl_2 + Cl_2$

It is because $Pb^{2+}$is more stable than Pb^{4+} due to inert pair effect.

- It is because both Pb and I are bigger in size Therefore, PbI_4 does not exist .secondly,I_2 is weaker oxidizing agent. $Pb^{4+}$is good oxidizing agent. Hence PbI_4 does not exist .

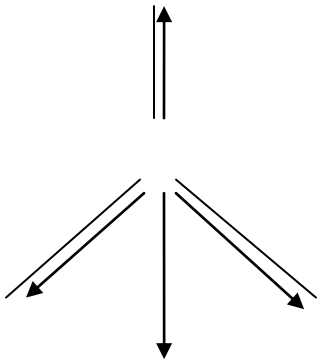
51.Suggest reasons why the B-F bond length in BF_3 (130 pm) and BF_4^- (143pm) differ.

Ans.In BF_3, B is sp^2 hybridized and its structure is trigonal planer whereas in BF_4^- , B is sp^3 hybridized and there is one coordinate covalent bond , therefore, bond length are different.

52.If B-Cl bond has a dipole moment,explain why BCl_3 molecule has zero dipole moment.

Ans.AlthoughB-Cl bonds are polar bonds they get cancelled due to trigonal planer shape ,net dipole moment is zero.

Cl



B 120^0

$\mu = 0$

ClCl

53.Aluminium trifluoride is insoluble in anhydrous HF but dissolves on addition of NaF. Aluminiumtrifluoride precipitates out of the resulting solution when gaseous BF_3 is bubbled through . Give reasons.

Ans.It is because HF is covalent compound and weak acid due to high bond dissociation energy whereas NaF is ionic compound.

$NaF + AlF_3 \longrightarrow Na_3[AlF_6]$

$Na_3[AlF_6] + 3BF_{3(g)} \longrightarrow AlF_3 + 3Na^+[BF_4]^-$

54. Suggest reasons as to why CO is poisonous.

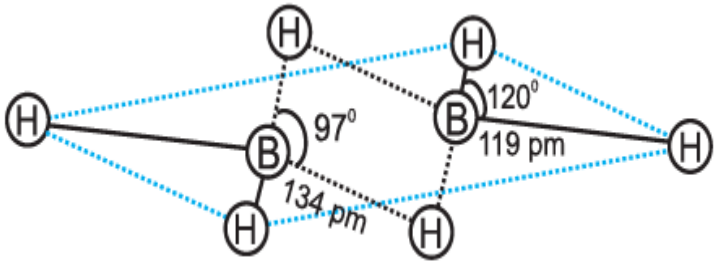
Ans.It is because CO react with haemoglobin to form carboxyhaemoglobin which does not act as oxygen carrier.

55.How is excessive content of CO_2 responsible for global warming ?

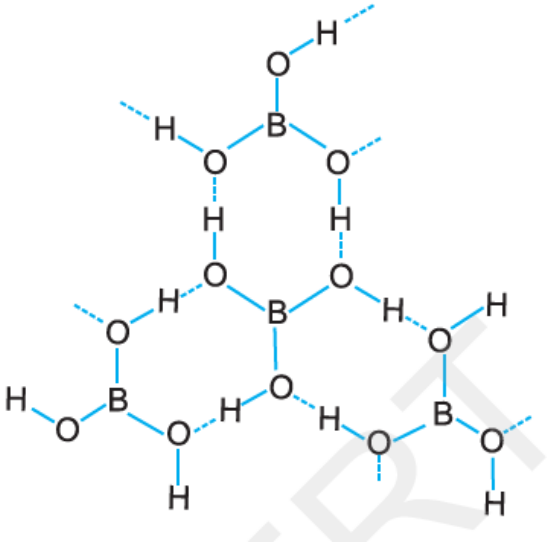
Ans.It is because CO_2 absorbs infra-red radiations which cause heating effect.

56. Explain structures of diboranes and boric acid

Ans.



(Structure of diborane)



(Structure of boric acid)

57.What happens when

(a)Borax is heated strongly.

(b) Boric acid is added to water .

(c) Aluminium is treated with dilute NaOH .

(d) BF_3 is reacted with ammonia .

Ans. $Na_2B_4O_7.10H_2O \xrightarrow{\triangle} Na_2B_4O_7 \xrightarrow{\triangle} NaBO_2 + B_2O_3$

$B(OH)_3 + H_2O \longrightarrow [B(OH)_4]^- + H^+$

$2Al + 2NaOH + 2H_2O \longrightarrow NaAlO_2 + 3H_2$

$BF_3 + NH_3 \longrightarrow [H_3NBF_3] \longrightarrow$

58.Explain the following reactions.

(a) Silicon is heated with methyl chloride at high temperature in the presence of copper .

(b) Silicon dioxide is treated with hydrogen fluoride.

(c) CO is heated with ZnO.

(d)Hydrated Alumina is treated with dilute NaOH .

Ans. $Si + 2CH_3Cl \xrightarrow[Heat]{Cu} Si(CH_3)_2Cl_2$

$SiO_2 + 6HF \longrightarrow H_2SiF_6 + 2H_2O$

$ZnO + CO \longrightarrow Zn + CO_2$

$Al_2O_3.2H_2O + 2NaOH \longrightarrow 2NaAlO_2 + 3H_2O$

59.Give reasons:

(a) Conc. HNO_3 can be transported in aluminium container.

(b) a mixture of dilute NaOH and aluminium pieces is used to open drain.

(c) Graphite is used as lubricant.

(d) Diamond is used as an abrasive.

(e) Aluminium alloys are used to make aircraft body.

(f) Aluminium utensils should not be kept in water overnight.

(g) Aluminium wire is used to make transmission cables.

Ans. (a) It is because aluminium container coated with Al_2O_3 which makes it passive and does not react with conc. HNO_3

(b) It is because it liberates hydrogen gas which exerts pressure to open drain.

(c) Graphite is soft and slippery and has high melting point, therefore , acts as good lubricant

(d) It is hard substance therefore, used as abrasive.

(e) It is light ,tough and strong.

(f) It forms layer of $Al(OH)_3$ on the surface.

(g) it is good conductor of electricity.

60. Explain why is there a phenomenal decrease in ionization enthalpy from carbon to silicon ?

Ans. It is due to increase in atomic size , therefore ionisation energy decreases .

61. How would you explain the lower atomic radius of Ga as compared to Al?

Ans.It is due to poor shielding effect of d-electrons in Ga.

62. What are allotropes ? Sketch the structure of two allotropes of carbon namely diamond and graphite .What is the impact of structure on physical properties of two allotropes?

Ans. Allotropy is property due to which an element exists in more than one forms which differ in physical properties but have similar chemical properties and these forms are called allotropes.

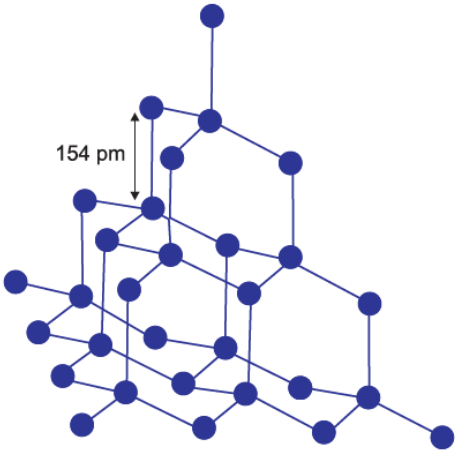


Fig. 11.3 The structure of diamond

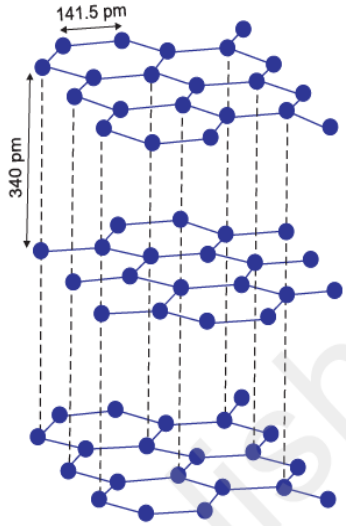


Fig 11.4 The structure of graphite

The structure of diamond makes it hard having high melting point and poor conductor of electricity.Structure of graphite makes it soft ,slippery and good conductor of electricity.

63. (a)Classify following oxides as neutral, acidic ,basic or amphoteric:

CO , B_2O_3 , SiO_2 , CO_2 , Al_2O_3 , PbO_2 , Tl_2O_3

(b)Write suitable chemical equation to show their nature.

Ans. (a)Neutral : CO

Acidic : B_2O_3 , CO_2, SiO_2,

Basic : Tl_2O_3

Amphoteric : Al_2O_3 , PbO_2

(b) CO is neither acidic or basic , it does not react with acid as well as base at room temperature.

$B_2O_3 + 6NaOH \longrightarrow 2Na_3BO_3 + 3H_2O$

$CO_2 + 2NaOH \longrightarrow Na_2CO_3 + H_2O$

$Tl_2O_3 + 6HCl \longrightarrow 2TlCl_3 + 3H_2O$

$SiO_2 + 2NaOH \longrightarrow Na_2SiO_3 + H_2O$

$Al_2O_3 + 2NaOH \longrightarrow 2NaAlO_2 + H_2O$

$Al_2O_3 + 2NaOH \longrightarrow 2AlCl_3 + 3H_2O$

$Al_2O_3 + 6HCl \longrightarrow 2AlCl_3 + 3H_2O$

$PbO_2 + 4HCl \longrightarrow PbCl_2 + Cl_2 + 2H_2O$

$PbO_2 + 2NaOH \longrightarrow Na_2PbO_3 + H_2O$

64. In some of the reaction thallium resembles aluminium , whereas in others it resembles with group 1 metals. Supports this statement by giving some evidences.

Ans. In addition +3, Tl also Shows +1 oxidation state due to inert pair effect . Thus it resembles with group 1 metals .Tl metals forms basic oxide like group 1 elements.

65.What mean by Catenation ? Why Does 'Carbon' Show the property of catenation to maximum extent?

Ans. Catenation is property due to which atoms of an element can perform stable covalent bonds with each other . Carbon shows the properties of catenation to maximum extent due to its smallest size, therefore , bond dissociation energy decreases.

66.Explain why BCl_3 is monomer but BH_3 exists as B_2H_6.

Ans In BCl_3 there is a back bonding i.e., Lone pair of electrons donated from chlorine to boron thus reducing its electron deficiency.

In BH_3 , back bonding is not possible but H-bridges are possible , therefore it exists as dimer.

67.Why is BF_3 planar molecule but NH_3 pyramidal ?

Ans In BF_3 (b) is sp^2 hybridised whereas in NH_3 , (N) is sp^3hybridised with one lone pair of electron , therefore it is pyramidel while BF_3 is planar.

68. How is diborane prepared in the laboratory ? Give its structure.

Ans. $2NaBH_4 + I_2 \xrightarrow{\text{diglyme}} 2NaI + B_2H_6 + H_2$

sodium borohydride (Diborane)

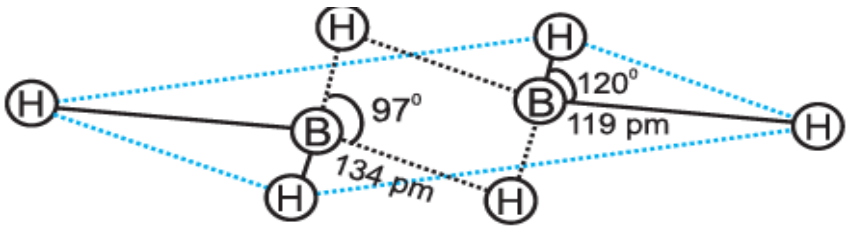


Fig.11.2(a) *The structure of diborane, B_2H_6*

69. Account the following :

Why BF_3 is less acidic than BCl_3 though fluorine is more electronegative than chlorine?

Ans.BF_3 is less acidic because back bonding is most effective due to 2p-2p overlapping whereas it is less effective in BCl_3 due to 2p-3p overlapping.

70. Give chemical reaction in support of each of the following statements:

(1) The +1 oxidation state gets stabilised progressively from Ga to Tl in group 13.

(2) All the bonds in PCl_5 molecule are not equivalent.

Ans. (1) It is due to inert pair effect which is due to poor shielding effect of d and f - electrons , as a result of which effective nuclear charge increases.

(2) it is because all the bonds are not in same plane. Axial bonds are longer than equatorial bonds due to more repulsion .

71. Give the chemical reaction as an evidence for each of the following observations:

(1) Tin (II) is a reducing agent whereas lead (II) is not.

(2) Gallium (I) undergoes disproportionation reaction.

Ans. It is because Pb^{2+} is more stable than Pb^{4+} due to inert pair effect whereas Sn^{4+} is more stable than Sn^{2+} therefore Sn^{2+} is a good reducing agent ,Pb^{2+} is not .

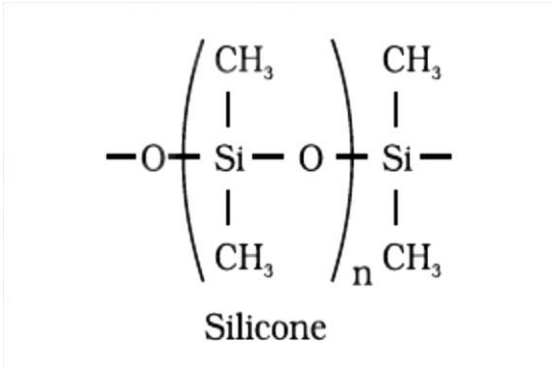
(2)It is because Ga^{3+} is more stable than Ga^{+} .

$3Ga^{+} \longrightarrow 2Ga + Ga^{3+}$

72. Write the chemical equation for the reaction which occur when :

R_2SiCl_2 is subjected to hydrolysis.

Ans . $nR_2SiCl_2 + nH_2O \longrightarrow (R_2SiO)_n$



Silicone

73. Explain why $PbCl_4$ is less stable than $SnCl_4$.

Ans. Pb^{4+} is less stable than Sn^{4+} due inert pair effect , Pb^{4+} gains 2 electrons to form Pb^{2+} .

$Pb^{4+} + 2e^{-} \longrightarrow Pb^{2+}$

Secondly , Sn^{4+} has greater effective nuclear charge than Pb^{4+} .

73. Complete the following chemical equation:

(1)RCl + Si $\xrightarrow[570k]{Cu}$

Ans. $2RCl + Si \xrightarrow[570k]{Cu} R_2SiCl_2$

74. Describe the two similarities and two dissimilarities between B and Al.

Ans . Similarities : (1) Both have similar electronic configuration.

(2) Both have same number of valence electron .

Dissimilarities : (1) B is non-metal ; Al is metal .

(2) B forms acidic oxide (B_2O_3) , whereas Al forms Al_2O_3 which is amphoteric .

75. Explain why BCl_3 is monomer but anhydrous $AlCl_3$ has dimeric structure ?

Ans. BCl_3 cannot exists as a dimer because it does not have d-orbitals whereas $AlCl_3$ can exist as a dimer because Al has d- orbital which can accept a pair of electrons from Cl to form coordinate bond .

76. Give a differences in structures of the following pair of compounds :

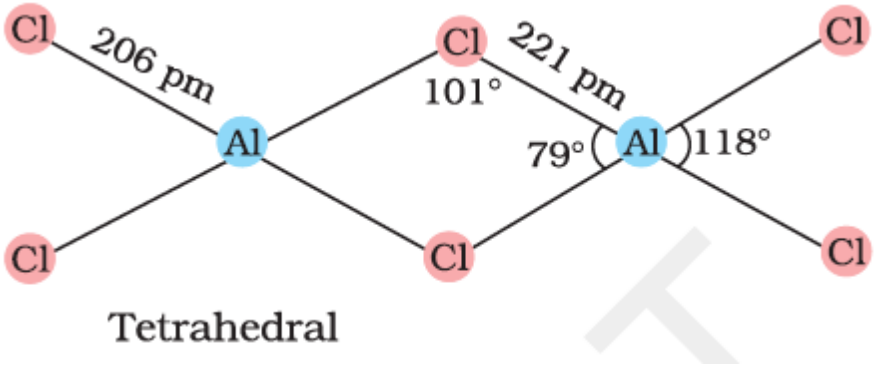
CO_2 and SiO_2 .

Ans. CO_2 is a linear molecule and exists as a monomer . It is gas while SiO_2 is solid at room temperature due to three dimensional network.

In each Si atom is covalently bonded to four oxygen atoms tetrahedrally. In CO_2 , 'C' is sp-hybridised while in SiO_2 , 'Si' is sp^3 hybridised .

77 . Draw a labelled diagram to show the dimeric structure of Al(III) chloride . state one use of anhydrous $AlCl_3$

Ans . The structure of $AlCl_3$ is shown below:



It is used in preparation of organic compounds in Friedal Crafts reaction.

78 . Arrange the following compounds in decreasing order of property indicated against each .

Give reason for your answer:

BCl_3 , $AlCl_3$, $GaCl_3$, $InCl_3$, (stability of +3 oxdation state)

Ans . BCl_3 > $AlCl_3$> $InCl_3$> $TiCl_3$ because of inert pair effect stability of +3

Oxidation state decreases .

79 . Discuss the diagonal relationship of Be and Al with regard to (i) the action of alkali and (ii) the structure of chlorides .

Ans . Be and Al resemble with each other due to same charge over radius ratio :

(i)Both Be and Al form amphoteric oxides .

(ii) Be and Al form covalent compounds .

(iii) $BeCl_2$ exists as polymer whereas $AlCl_3$ exists as dimer .

(iv) Both Be and Al react with NaOH to form similar compounds .

81 . C and Si are always tetravalent Ge , Sn , Pb show divalency .

Ans . Ge , Sn , Pb show divalency due to inert pair effect , Pb^{2+} is more stable than Pb^{4+}.

82 . Which of of the following is acidic and why?

SiO_2 , Al_2O_3 , PbO_2 , SnO_2

Ans. SiO_2 is acidic oxide because Si is non-metal .

83 . As we move down in group 13 elements increases in atomic size is comparatively very less . Explain .

Ans . It is due to poor Shielding effect of d and f-electrons , as a result of which effective nuclear charge increases .

84 . Gallium has high I.E than Al. Explain

Ans . It is due to poor Shielding effect of d-electrons in Ga , effective nuclear charge increases as compared to Al , therefore , I.E of Ga is higher than Al , Secondly Ga is smaller than Al .

85 . Boron does not form $(BF_6)^{3-}$ whereas $(AlF_6)^{3-}$ exists why ?

Ans . It is because 'B' does not have vacant d – orbitals whereas Al has vacant d – orbitals .

86 . Write the chemistry involved in etching of glass .

Ans .

$SiO_2 + 4HF \longrightarrow SiF_4 + H_2O$

$SiF_4 + 2HF \longrightarrow H_2SiF_6$

83 . Contrast the structures and properties of CO_2 and SiO_2 .

Ans . CO_2 is gas , Whereas SiO_2 is solid

CO_2 exists as descrete molecules whereas SiO_2 is three dimensional solid.

84 . Elemental Silicon does not form graphite like structure as carbon does . Give reason :

Ans . It is because 'Si' cannot form pπ- pπ bond like carbon in graphite is due to larger atomic size .

85 . What are silicones ? How are they manufactured ?

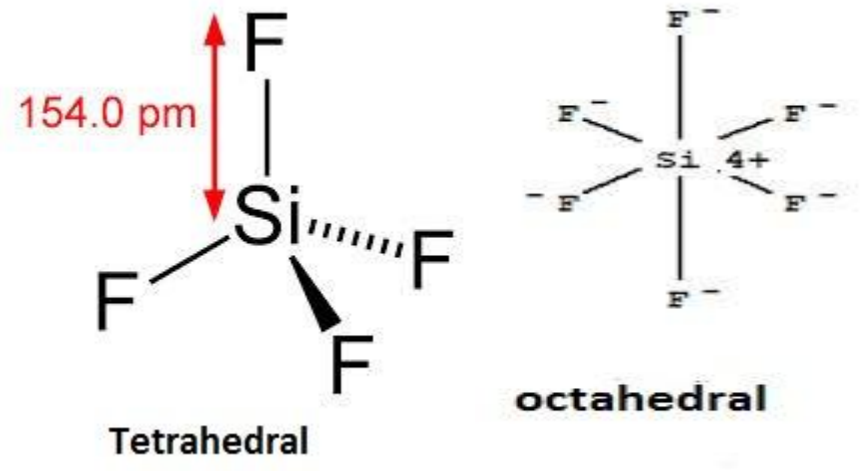
Ans . Silicones are rubber like polymers which are heat resistant and water repellant . They are prepared by Hydrolysis of chlorosilanes.

$2RCl + Si \xrightarrow[570K]{Cu} R_2SiCl_2$

$nR_2SiCl_2 + nH_2O \longrightarrow$

86 . Give the shapes of (a) SiF_4 (b) SiF_6^{2-} .

Ans.



86. Arrange the following in decreasing order of boiling point giving reason:

CH_4 , SiH_4 , GeH_4 , SnH_4 .

Ans. $SnH_4 > GeH_4 > SiH_4 > CH_4$

As we go down the group from C to Sn surface area of hydrates

increases , therefore , vanderwaals" force increases, hence boiling point increases.

87. Explain the following :

(a) CaO reacts with SiO_2 to form a slag

(b) aluminium oxides cannot be reduce by carbon.

Ans. (a) $CaO + SiO_2 \longrightarrow CaSiO_3$ (Slag)

Basic oxide Acidic oxide (Calcium silicate)

(Flux) (Gangue)

(b) It is because 'Al' is stronger reducing agent than carbon .

88 . (a) What is inorganic Benzene ?

(b) State the types of hybridization of carbon in diamond and in graphite .

Ans. (a) $B_3N_3H_6$ is inorganic benzene.

(b) 'C" in diamond is sp^3 hybridised whereas in graphite 'C' is sp^2 hybridised.

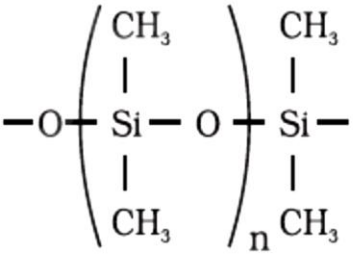
89 . How is Lithium Aluminium hydride prepared ? Write its one of the most important use.

Ans. $4LiH + AlCl_3 \longrightarrow LiAlH_4 + 3LiCl$

It is used as reducing agent.

90. If the starting material for the manufacture of silicones is $RSiCl_3$, Write the structure of the product formed .

Ans. $nRSiCl_3 + nH_2O \longrightarrow (R_2SiO)_n$



Silicone

91 . How far do you agree with the phenomenon concept of 'inert pair effect' ?Justify your answer.

Ans. Inert pair effect is not very accurate concept . The better way of understanding is that energy required to unpair two electrons in s – orbital is higher than energy released in formation of two additional bonds .

92 . When metal (X) is treated with sodium hydroxide a white precipitate (A) is obtained , which is soluble in excess of NaOH to give soluble complex (B) . compound (A) is soluble in dilute HCl to form compound (C) . The compound (A) when heated strongly gives (D) , which is used to extract metal . Identify (X) ,(A) , (B) , (C) , and (D) . Write suitable equations to support their identities .

Ans. $Al + 3NaOH \longrightarrow Al(OH)_3 + 3Na$

'X' 'A'

$Al(OH)_3 + NaOH \longrightarrow Na[Al(OH)_4]$

soluble 'B'

$Al(OH)_3 + 3HCl \longrightarrow AlCl_3 + 3H_2O$

'C'

$2Al(OH)_3 \longrightarrow Al_2O_3 + 3H_2O$

'D'

"D" is used to extract metal 'X'

93. What do you understand by (a) inert pair effect (b) Allotropy and (c) Catenation ?

Ans. (a) Inert pair effect : The pair of electrons in valence s –orbitals does not take part in bond formation , it is called inert pair effect.

(b) Allotropy: It is property due to which an element exists in two or more forms which differ in physical properties but have similar chemical properties.

(c) Catenation : It is property is due to whidh an element forms stable covalent bonds with atom of itself.

94. What is dry ice ? why is it called so how will you prepare pure sample of CO(carbon mono oxide).

Ans. Solid CO_2 is called dry ice . It is called dry ice Because it directly change into gaseous state without becoming liquid .

$$HCOOH \xrightarrow[373k\text{pure}]{conc.H_2SO_4} CO + H_2O$$

95 . Write balance equation for the following reaction and name the main product formed in each case .

(i) $NaBH_4 + I_2$ ⟶

(ii) $B_2H_6 + NaH$ ⟶

(iii) $BF_3 + LiH$ $\xrightarrow{450k}$

(iv) $SiCl_4 + H_2O$ ⟶

Ans. (i) $2NaBH_4 + I_2 \longrightarrow 2NaI + \underset{\text{Diborane}}{B_2H_6} + H_2$

(ii) $B_2H_6 + 2NaH \longrightarrow \underset{\text{Sodium borohydride}}{2NaBH_4}$

(iii) $2BF_3 + 6LiH \xrightarrow{450k} \underset{\text{Diborane}}{B_2H_6} + 6LiF$

(iv) $\underset{\text{Silicic acid}}{SiCl_4 + 2H_2O} \longrightarrow Si(OH)_4 + 4HCl$

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